

Intergenerational Housing Allocation and Fertility

Part I: Compact Analytical Model

Tommaso De Santo

June 2026

Question: when does limited access to family-capable housing reduce fertility, and when is policy effective?

1. Primitives and Timing

Households. A mass $\bar{M}$ of potential young households has type i , drawn from G_Y . A type carries income y_i , pledgeable wealth a_i , inheritance exposure B_i , and child consumption cost χ_i . Old incumbent owners are indexed by j , drawn from F_O .

Timing for young households.

1. Decide whether to enter the modeled housing market.
2. Conditional on entry, solve renter and owner branch problems.
3. Choose tenure $o_i \in \{R, O\}$, housing h_i , residual adult consumption c_i , and fertility n_i .

Child inputs.

$$c_i = C_i - \chi_i n_i, \quad s_i = h_i - \kappa n_i,$$

$$u_i^Y(c_i, h_i, n_i) = \log c_i + \alpha \log(h_i - \kappa n_i) + \beta \log n_i.$$

α is the adult housing-services weight, β is the fertility weight, and κ is housing services per child.

2. Housing Segments, Prices, and Supply

Two housing segments.

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \quad 0 < \bar{h}^R < \bar{h}^O.$$

$\bar{h}^m$ is an individual occupancy cap. Aggregate stock is H^m .

Stocks and construction.

$$H^m = \bar{H}^m + I^m, \quad q^m = \Phi_l^m(I^m; Z^m), \quad m \in \{R, O\}.$$

$\bar{H}^m$ is inherited stock, I^m is net construction, Φ^m is the flow resource cost, and Z^m shifts construction ease.

Owner stock accounting.

$$\int H_j^0 dF_O(j) + H_{\text{pass}}^O = \bar{H}^O.$$

Old owners enter with endowments H_j^0 ; remaining owner stock is held by a passive account.

Owner price.

$$P^O = \frac{q^O}{r + \delta^O + \tau^P}.$$

q^O is the flow user cost; P^O is the asset price used in down-payment constraints. r is the interest rate, δ^O depreciation, and τ^P the property-tax rate.

3. Young Household Problem

Pledgeable resources.

$$x = 1 - \tau^b, \quad A_i(x) = a_i + xB_i + T_i^b(x).$$

$A_i(x)$ enters purchase constraints, not the flow budget.

Renter branch.

$$W_i^R(q^R) = \max_{c,h,n} u_i^Y(c, h, n)$$

$$c + q^R h + \chi_i n = y_i, \quad h \leq \bar{h}^R, \quad h - \kappa n > 0.$$

Owner branch.

$$W_i^O(q^O, P^O, x) = \max_{c,h,n} u_i^Y(c, h, n)$$

$$c + q^O h + \chi_i n = y_i, \quad h \leq \bar{h}^O, \quad (1 - \phi)P^O h \leq A_i(x), \quad q^O h \leq \psi y_i.$$

ϕ is the financed share; ψ is the payment-to-income limit.

4. Tenure, Entry, and Old Owners

Tenure choice conditional on entry.

$$W_i^Y(q^R, q^O, P^O, x) = \max\{W_i^R(q^R), W_i^O(q^O, P^O, x)\}.$$

Logit entry.

$$\pi_i^E = \frac{\exp\{W_i^Y/\kappa_E\}}{\exp\{W_i^Y/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}.$$

$\bar{W}^E$ is the outside option and κ_E is the entry taste-shock scale.

Old incumbent owners. Old owner j enters with owner housing endowment H_j^0 . If she keeps the current house, she pays tax on a protected assessment. The per-unit tax discount from staying put is

$$L_j^\tau = \tau^P P^O - \tilde{\tau}_j \tilde{P}_j \geq 0.$$

Here $\tau^P P^O$ is the tax cost at current market value and $\tilde{\tau}_j \tilde{P}_j$ is the protected tax cost. Old owner j chooses how much current housing to keep, consumption, and bequest expenditure:

$$\max_{c, h, e} u_j^O(c, h) + \mathcal{B}_j(e) \quad \text{s.t.} \quad c + e + (q^O - L_j^\tau)h = m_j + q^O H_j^0, \quad 0 \leq h \leq H_j^0.$$

m_j is nonhousing resources. She feels carrying cost $q^O - L_j^\tau$:

$$MV_j^O = q^O - L_j^\tau.$$

5. Competitive Equilibrium

Given policies and stocks, an equilibrium is

$$(q^R, q^O, P^O, I^R, I^O, \{\pi_i^E, o_i, c_i, h_i, n_i\}_i, \{c_j^O, h_j^O, e_j\}_j)$$

such that:

1. Household optimization. Young households solve renter and owner branches, choose tenure by $W_i^Y = \max\{W_i^R, W_i^O\}$, and enter with probability π_i^E . Old owners choose consumption, bequests, and how much current housing to keep.

2. Construction and asset pricing.

$$q^R = \Phi_I^R(I^R; Z^R), \quad q^O = \Phi_I^O(I^O; Z^O), \quad P^O = \frac{q^O}{r + \delta^O + \tau^P}.$$

3. Segment clearing.

$$\bar{M} \int \pi_i^E \mathbf{1}\{o_i = R\} h_i^R dG_Y(i) = \bar{H}^R + I^R,$$
$$\bar{M} \int \pi_i^E \mathbf{1}\{o_i = O\} h_i^O dG_Y(i) + \int h_j^O dF_O(j) = \bar{H}^O + I^O.$$

4. Accounts. Fiscal revenues, passive housing income, and policy-account costs are rebated through the transfer rules already embedded in $A_i(x)$, y_i , and old resources.

6. Planner Problem

Choices. The planner chooses tenure assignments o_i^P , young allocations (c_i, h_i, n_i) , entry values π_i^P , old allocations (c_j^O, h_j^O, e_j) , construction (I^R, I^O) , and policy instruments.

Segment feasibility.

$$\bar{M} \int \pi_i^P \mathbf{1}\{o_i^P = R\} h_i dG_Y(i) \leq \bar{H}^R + I^R,$$
$$\bar{M} \int \pi_i^P \mathbf{1}\{o_i^P = O\} h_i dG_Y(i) + \int h_j^O dF_O(j) \leq \bar{H}^O + I^O.$$

Goods feasibility.

$$\bar{M} \int \pi_i^P (c_i + \chi_i n_i - y_i) dG_Y(i) + \int (c_j^O + e_j - m_j) dF_O(j) + \Phi^R + \Phi^O + \Xi^\Pi \leq 0.$$

Implementation. Menus $h_i \in \mathcal{H}^m$ are real constraints. Down-payment and payment-to-income constraints remain unless the planner uses feasible instruments whose resource costs enter Ξ^Π .

7. Planner Marginal Conditions

Segment shadow prices. Let ΛQ^R and ΛQ^O be the multipliers on rental and owner segment feasibility; Λ is the goods multiplier.

Interior allocation rules away from real menu corners.

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = Q^R \quad \text{for renter assignments,} \quad \frac{u_{h,i}^Y}{u_{c,i}^Y} = Q^O \quad \text{for young owner assignments,}$$

$$\frac{u_{h,j}^O}{u_{c,j}^O} = Q^O \quad \text{for old owner housing,} \quad Q^m = \Phi_I^m(I^m; Z^m).$$

Fertility rule.

$$\frac{\beta c_i}{n_i} + \nu_i^g = \chi_i + \kappa Q_i^{oP}.$$

ν_i^g is the goods-equivalent social value of births beyond parental utility; in the fertility-neutral planner, $\nu_i^g = 0$.

Connection to CE prices. At an interior competitive allocation with construction and segment clearing,

$$Q^R = q^R, \quad Q^O = q^O.$$

8. Where the CE Differs from the Planner

Young owner financial constraints. If down-payment or payment-to-income constraints bind, the competitive owner margin is

$$MV_i^O = q^O + \zeta_i^{O,F}, \quad \zeta_i^{O,F} = \frac{\mu_i}{\lambda_i^O} (1 - \phi) P^O + \frac{\eta_i}{\lambda_i^O} q^O.$$

$$\underbrace{MV_i^O}_{\text{young private value}} - \underbrace{Q^O}_{\text{planner scarcity cost}} = \underbrace{\zeta_i^{O,F}}_{\text{financial access gap}} \quad (Q^O = q^O).$$

λ_i^O is the owner budget multiplier; μ_i and η_i are down-payment and payment-to-income multipliers.

Old owners: incumbent tax discount.

$$MV_j^O = q^O - L_j^\tau, \quad \underbrace{Q^O}_{\text{planner scarcity cost}} - \underbrace{MV_j^O}_{\text{old private value}} = \underbrace{L_j^\tau}_{\text{tax discount from staying put}}.$$

Efficiency statement. With zero marginal resource cost of relaxing DP/PTI, positive $\zeta_i^{O,F}$ or L_j^τ violates the planner's marginal conditions. With positive instrument cost, compare the gap to that cost.

9. Fertility Effect of a Family-Space Cap

Effective cap in each branch.

$$H_i^R = \bar{h}^R, \quad H_i^O = \min \left\{ \bar{h}^O, \frac{A_i(x)\rho_p}{(1-\phi)q^O}, \frac{\psi y_i}{q^O} \right\}, \quad \rho_p = r + \delta^O + \tau^p.$$

Within-branch response. For a fixed tenure branch with user cost q , effective cap H , and residual consumption and housing (c, s) ,

$$\Psi_i^m \equiv \frac{\partial n_i^m}{\partial H_i^m} = \frac{\alpha\kappa/s^2 - \chi_i q^m/c^2}{\beta/n^2 + \chi_i^2/c^2 + \alpha\kappa^2/s^2}.$$

Sign. If the child input bundle is not more consumption-intensive than the adult residual bundle,

$$\chi_i \leq \frac{\kappa q^m}{\alpha},$$

then relaxing a strictly binding family-housing cap raises fertility within the fixed branch.

10. Local Policy Decomposition

Objects. For policy z , let $P_i^m = \partial H_i^m / \partial z$, $\Theta_i^m = \lambda_i^m \zeta_i^m$, and $\mathcal{B}_m$ be types whose chosen branch m has a binding effective cap.

Fixed-price, fixed-tenure, fixed-active-set formula.

$$\begin{aligned} \frac{dN}{dz} \Big|_{q,o,\mathcal{A}} &= \bar{M} \sum_m \int_{\mathcal{B}_m} \underbrace{\pi_i^E}_{\text{entered mass}} \underbrace{\Psi_i^m}_{\text{fertility response}} \underbrace{P_i^m}_{\text{policy pass-through to } H_i^m} dG_Y(i) \\ &+ \bar{M} \sum_m \int_{\mathcal{B}_m} \underbrace{(n_i^m - \bar{n}_i^E)}_{\text{births relative to outside option}} \underbrace{\frac{\pi_i^E(1 - \pi_i^E)}{\kappa_E}}_{\text{entry sensitivity}} \underbrace{\Theta_i^m}_{\text{utility gain policy from } H_i^m} \underbrace{P_i^m}_{\text{policy pass-through to } H_i^m} dG_Y(i). \end{aligned}$$

Reading it. The first line is the intensive fertility response of already-entered constrained households. The second line is the entry response. Tenure switching, price feedback, construction responses, and resource costs Ξ^Π are outside this local formula.

Appendix: Branch Derivative

If $0 < H < h^u$, then $h(H) = H$ and

$$c(H) = y - qH - \chi n(H), \quad s(H) = H - \kappa n(H).$$

The constrained fertility condition is

$$F(n, H) \equiv \frac{\beta}{n} - \frac{\chi}{c(H)} - \frac{\alpha\kappa}{s(H)} = 0.$$

$$F_n = - \left[\frac{\beta}{n^2} + \frac{\chi^2}{c^2} + \frac{\alpha\kappa^2}{s^2} \right] < 0, \quad F_H = \frac{\alpha\kappa}{s^2} - \frac{\chi q}{c^2}.$$

$$\frac{dn}{dH} = - \frac{F_H}{F_n} = \frac{\frac{\alpha\kappa}{s^2} - \frac{\chi q}{c^2}}{\frac{\beta}{n^2} + \frac{\chi^2}{c^2} + \frac{\alpha\kappa^2}{s^2}}.$$

Appendix: Equal Scaling and Access Gaps

At a binding cap, the household marginal value of housing is

$$\frac{u_h}{u_c} = \frac{\alpha c_i^m}{s_i^m} = q^m + \zeta_i^m.$$

Under exact equal scaling, $\chi_i = \kappa q^m / \alpha$. The numerator of Ψ_i^m is

$$\frac{\alpha \kappa}{(s_i^m)^2} - \frac{\chi_i q^m}{(c_i^m)^2} = \frac{\kappa}{\alpha} \left(\frac{\alpha}{s_i^m} - \frac{q^m}{c_i^m} \right) \left(\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m} \right).$$

Since

$$\frac{\alpha}{s_i^m} - \frac{q^m}{c_i^m} = \frac{\zeta_i^m}{c_i^m},$$
$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m} \Omega_i^m, \quad \Omega_i^m > 0.$$

Appendix: Planner Comparison

At an interior competitive allocation with construction,

$$Q^R = q^R = \Phi_I^R(I^R; Z^R), \quad Q^O = q^O = \Phi_I^O(I^O; Z^O).$$

Planner housing rules require

$$MV_i^R = q^R, \quad MV_i^O = q^O, \quad MV_j^O = q^O.$$

Competitive marginal values are

$$MV_i^R = q^R + \zeta_i^R, \quad MV_i^O = q^O + \zeta_i^O, \quad MV_j^O = q^O - L_j^\tau.$$

When real menu corners are common to households and planner, the implementation-relevant young gap is the financial component:

$$\zeta_i^{O,F} = \frac{\mu_i}{\lambda_i^O} (1 - \phi) P^O + \frac{\eta_i}{\lambda_i^O} q^O.$$

If financial constraints are relaxable at zero marginal real resource cost, $\zeta_i^{O,F} > 0$ violates the planner's marginal condition.